The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 October 2000.

INCH-POUND

MIL-PRF-19500/290H
10 July 2000
SUPERSEDING
MIL-PRF-19500/290G
27 August 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPES 2N2904, 2N2904A, 2N2904AL, 2N2905, 2N2905A, AND 2N2905AL, JAN, JANTX, JANTXV, JANJ, JANS, JANTXVM, JANTXVD, JANTXVR, JANTXVH, JANSM, JANSD, JANSR, AND JANSH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, switching transistors. Five levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Provision for radiation hardness assurance (RHA) to four radiation test levels is provided for JANTXV and JANS product assurance levels for type 2N2905A. RHA level designators "M", "D", "R", and "H" are appended to the device prefix to identify devices which have passed RHA requirements.

1.2 Physical dimensions. See figure 1 herein (similar to TO-39).

1.3 Maximum ratings.

Types	P_T = (1) T_A = +25°C	P_T = (2) T_C= +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{STG} and T_J
	W	W	V dc	V dc	V dc	mA dc	°C
2N2904, 2N2905	0.8	3.0	60	40	5	600	-65 to +200
2N2904A, 2N2905A, 2N2904AL, 2N2905AL	0.8	3.0	60	60	5	600	-65 to +200

(1) Derate linearly, 5.7 mW/°C for T_A > +60°C.
(2) Derate linearly, 17.2 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics at T_A = +25°C.

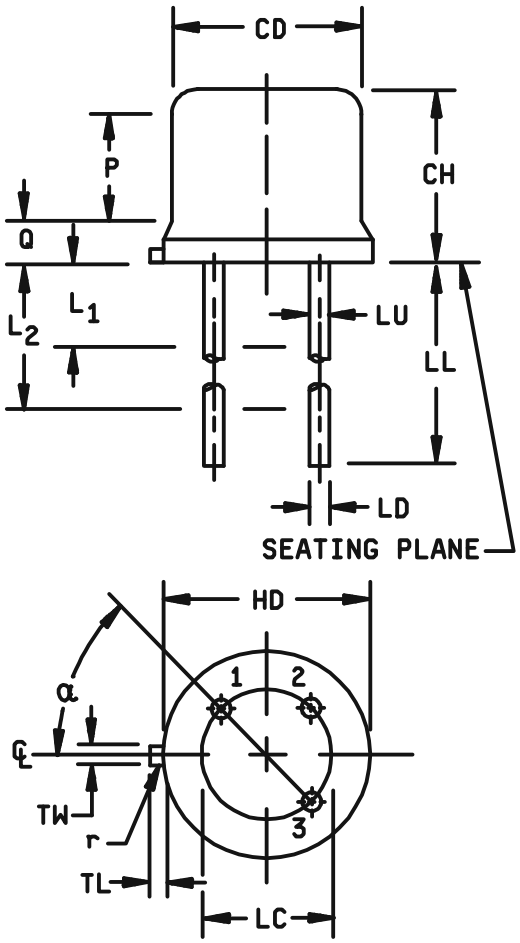
	h_{FE} at V_{CE} = 10 V dc					$R_{\theta JA}$
	h_{FE1} I_C = 0.1 mA dc	h_{FE2} I_C = 1 mA dc	h_{FE3} I_C = 10 mA dc	h_{FE4} (1) I_C = 150 mA dc	h_{FE5} (1) I_C = 500 mA dc	°C/mW max
	2N2904	2N2904	2N2904	2N2904	2N2904	
Min	20	25	35	40	20	
Max	---	175	---	120	---	0.29
	2N2905	2N2905	2N2905	2N2905	2N2905	
Min	35	50	75	100	30	
Max	---	450	---	300	---	0.29
	2N2904A 2N2904AL	2N2904A 2N2904AL	2N2904A 2N2904AL	2N2904A 2N2904AL	2N2904A 2N2904AL	
Min	40	40	40	40	40	
Max	---	175	---	120	---	0.29
	2N2905A 2N2905AL	2N2905A 2N2905AL	2N2905A 2N2905AL	2N2905A 2N2905AL	2N2905A 2N2905AL	
Min	75	100	100	100	50	
Max	---	450	---	300	---	0.29

(1) Pulsed (see 4.5.1).

Types 2N2904 2N2904A 2N2904AL 2N2905 2N2905A 2N2905AL	$\|h_{fe}\|$ f = 100 MHz V_{CE} = 20 V dc I_C = 50 mA dc	C_{obo} 100 kHz ≤ f ≤ 1 MHz V_{CB} = 10 V dc I_E = 0	Switching	
			t_{on} (See figure 2)	t_{off} (See figure 3)
		pF	ns	ns
Min	2.0	---	---	---
Max	---	8	45	300

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	0.305	0.335	7.75	8.51	
CH	0.240	0.260	6.10	6.60	
HD	0.335	0.370	8.51	9.40	
LC	0.200 TP		5.08 TP		6
LD	0.016	0.021	0.41	0.53	7, 8
LL	0.500	0.750	12.70	19.05	7, 8, 12
LU	0.016	0.019	.041	0.48	7, 8
L_1	---	0.050	---	1.27	7, 8
L_2	0.250	---	6.35	---	7, 8
Q	---	0.050	---	1.27	5
TL	0.029	0.045	0.74	1.14	4
TW	0.028	0.034	0.71	0.86	3
r	---	0.010	---	0.25	10
α	45° TP		45° TP		6
P	0.100		2.54		



NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of 0.011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane 0.054 +0.001, -.000 inch (1.37 +0.03, -0.00 mm) below seating plane shall be within 0.007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ∅x symbology.
12. For "L" suffix devices, dimension LL is 1.50 (38.10 mm) minimum, 1.75 (19.05 mm) maximum.
13. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (similar to TO-39).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking may be omitted from the body, but shall be retained on the initial container.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and Table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in Table I, group A herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANJ qualification. For JANJ level, 3.3.1 through 3.3.1.3 of MIL-PRF-19500 shall apply, except as modified herein. Supplier imposed requirements as well as alternate screens, procedures, and/or process controls shall be documented in the quality management (QM) plan and must be submitted to the qualifying activity for approval. When alternate screens, procedures, and/or process controls are used in lieu of the JANJ screens herein, equivalency shall be proven and documented in the QM plan. Radiation characterization may be submitted in the QM plan at the option of the manufacturer, however, paragraph 3.3.1.1 of MIL-PRF-19500 is not required. Lot formation and conformance inspection requirements for JANJ shall be those used for JANTXV devices as a minimum. Die lot controls and rework requirements shall be in accordance with MIL-PRF-19500 paragraphs 3.13 and D.3.13.2.1.

4.2.1.1 JANJ qualification. For JANJ qualification, 4.4.2.2 herein shall be performed as required by the qualifying activity.

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4.3 Screening (JANTX, JANTXV, JANJ, and JANS levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with Table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANJ level
1b	Required	Required
2	Optional	Optional
3a 3b 3c (1)	Required Not applicable Required (see 4.3.2)	Required Not applicable Required (see 4.3.2)
4	Required	Optional
5	Required	Required (one pass minimum)
7 (2)	Required	Required
8	Required	Not required (Bin and cell)
9	I_{CBO2}, h_{FE2}	I_{CBO2}, h_{FE2}
10	24 hours minimum	24 hours minimum
11	I_{CBO2}; h_{FE2}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ± 15 percent.	I_{CBO2}; h_{FE2}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ± 15 percent.
12	See 4.3.1, 240 hours minimum	See 4.3.1, 160 hours minimum
13 (3)	Subgroups 2 and 3 of table I herein; ΔI_{CBO2}= 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ± 15 percent.	Subgroups 2 and 3 of table I herein; ΔI_{CBO2}= 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ± 15 percent.
14a, and 14b (2)	Optional	Optional
14a 14b	Not applicable Optional	Not applicable Optional
15	Required	Required - Attributes data only, film or non-film techniques may be utilized.
16	Required	Required

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4.3 Screening (JANTX, JANTXV, JANJ, and JANS levels only) - Continued.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTXV and JANTX level
1b	Required (JANTXV only)
2	Not required
3a	Required
3b	Not applicable
3c (1)	Required (see 4.3.2)
4	Optional
5 and 6	Not applicable
7a and 7b (2)	Required
8	Not required
9	Not applicable
10	24 hours minimum
11	I_{CBO2}; h_{FE4}
12	See 4.3.1, 80 hours minimum
13	Subgroup 2 of table I herein; ΔI_{CBO2}= 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE4} = ± 15 percent.
14a and 14b	Optional
15 and 16	Not required

(1) Thermal impedance may be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) Hermiticy evaluation may be performed in either step 7 or step 14.
(3) PDA = 5 percent for screen 13, applies to ΔI_{CBO2}, Δh_{FE4}, I_{CBO2}, and h_{FE4}.

Thermal impedance ($Z_{\theta JX}$) is not required in screen 13.

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4.3.1. Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 Vdc

T_A = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5. Power shall be applied to the device to achieve a Junction temperature, T_J =135°C minimum using a minimum P_D = 75 percent of max P_T as defined in 1.3 herein.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3131.

a. I_M measurement current ..... 5 mA.

b. I_H forward heating current ..... 200 mA (minimum).

c. t_H heating time ..... 25 - 30 ms.

d. t_{md} measurement delay time ..... 60 µs maximum.

e. V_{CE} collector-emitter voltage ..... 10 V dc

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and group A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2 herein).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein. Group A inspection for JANJ shall be in accordance with JANTXV of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 - 30 V dc.
B5	1027	V_{CB} = 10 - 30 V dc; T_A = +125°C ± 25°C for 96 hours with P_T adjusted according to the chosen T_A to give T_J = +275°C minimum. Optionally this test may be performed for 216 hours minimum with P_T and T_A adjusted to achieve a T_J = +225°C minimum. A minimum P_D = 75 percent of rated P_T shall be applied. (Note: If a failure occurs, resubmission shall be at the test conditions of the original sample.)

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4.4.2.2 Group B inspection (JAN, JANTX, JANTXV and JANJ). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new "assembly lot" option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 - 30 V d, power shall be applied to achieve a junction temperature of T_J = +150°C minimum. A minimum of 75 percent of rated power shall be dissipated. No heat sink or forced- air cooling on devices shall be permitted. n = 45, C = 0.
2	1039	The steady-state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B step 2 shall not be required more than once for any single wafer lot. n = 45, C = 0.
3	1032	High- temperature life (non operating), T_A = +200°C, n = 22, C = 0, t = 340 hours

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements.:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, JANTXV and JANJ) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, (not applicable for UA and UB devices).
C6	1027	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 - 30 V d, Power shall be applied to achieve a junction temperature of T_J = +150°C minimum. A minimum of 75 percent of rated power P_T shall be dissipated. No heat sink or forced- air cooling on devices shall be permitted.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, JANTXV and JANJ) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UA and UB devices.
C6		Not applicable.

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4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Quality conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in Table II. These tests shall be performed as required in accordance with MIL-PRF-19500 and MIL-STD-750, method 1019 for total ionizing dose or method 1017 for neutron fluence as applicable.

4.4.5 Group E inspection. Group E inspection shall be performed for qualification or requalification only. The tests specified in Table III herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit
		Method	Conditions		
1.	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 50 V dc	ΔI_{CB02} (1)	100% of initial value or ± 8 nA dc, whichever is greater.
2.	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc; pulsed see 4.5.1	Δh_{FE2} (1)	± 25% change from initial reading.

(1) Devices which exceed the group A limits for this test shall not be accepted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical inspection 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Group A, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = + 250°C at t = 24 hours or T_A = + 300°C at t = 2 hours, n = 11 wires, c = 0				
Subgroup 2						
Collector to base, cutoff current	3036	V_{CB} = 60 V dc	I_{CBO1}		10	µA dc
Emitter to base, cutoff current	3061	V_{BE} = 5 V dc	I_{EBO1}		10	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2904, 2N2905 2N2904A, 2N2905A 2N2904AL, 2N2905AL				40 60 60		V dc V dc Vdc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO2}			
2N2904, 2N2905 2N2904A, 2N2905A 2N2904AL, 2N2905AL					20 10 10	nA dc nA dc nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 3.5 V dc	I_{EBO2}		50	nA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter cutoff current	3041	Bias condition C	I_{CES}		1	µA dc
2N2904, 2N2905		V_{CE} = 40 V dc				
2N2904A, 2N2904AL		V_{CE} = 60 V dc				
2N2905A, 2N2905AL		V_{CE} = 60 V dc				
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 0.1 mA dc	h_{FE1}			
2N2904				20		
2N2905				35		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				75		
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE2}			
2N2904				25	175	
2N2905				50	450	
2N2904A, 2N2904AL				40	175	
2N2905A, 2N2905AL				100	450	
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc	h_{FE3}			
2N2904				35		
2N2905				75		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				100		
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, pulsed (see 4.5.1)	h_{FE4}			
2N2904				40	120	
2N2904A, 2N2904AL						
2N2905				100	300	
2N2905A, 2N2905AL						
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE5}			
2N2904				20		
2N2905				30		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				50		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.4	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.6	V dc
Base emitter voltage (saturated)	3066	Test condition A, I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.3	V dc
Base emitter voltage (saturated)	3066	Test condition A, I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.6	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 50 V dc	I_{CBO3}			
2N2904, 2N2905					20	µA dc
2N2904A, 2N2904AL					10	µA dc
2N2905A, 2N2905AL					10	µA dc
Low temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE6}			
2N2904				15		
2N2905				30		
2N2904A, 2N2904AL				20		
2N2905A, 2N2905AL				50		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 1 mA dc, f = 1 kHz	h_{fe}			
2N2904				25		
2N2905				50		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				100		
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 20 V dc, I_C = 50 mA dc, f = 100 MHz	$\|h_{fe}\|$	2.0		
Open circuit output Capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 2.0 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz (see 4.5.2)	C_{ibo}		30	pF
Turn-on time		(See figure 2)	t_{on}		45	ns
Turn-off time		(See figure 3)	t_{off}		300	ns
Subgroups 5 and 6						
Not applicable						
Subgroups 7						
Decap internal visual (design verification)	2075	n = 1 device, c = 0				

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.

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TABLE II. Group D inspection and end-point limits.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Neutron irradiation	1017					
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}			
M2N2905A, D2N2905A, R2N2905A, H2N2905A		VCB = 50 V dc			10	nA dc
Forward current transfer Ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}			
M2N2905A				61		
D2N2905A				59		
R2N2905A				31		
H2N2905A				5		
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE2}			
M2N2905A				89	450	
D2N2905A				82	450	
R2N2905A				53	450	
H2N2905A				12	450	
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE3}			
M2N2905A				90		
D2N2905A				87		
R2N2905A				81		
H2N2905A				22		
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc	h_{FE4}			
M2N2905A				84		
D2N2905A				80		
R2N2905A				58		
H2N2905A				28		
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 50 mA dc	h_{FE5}			
M2N2905A				45		
D2N2905A				45		
R2N2905A				40		
H2N2905A				20		

See footnote at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 - Continued						
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$			
M2N2905A, D2N2905A, R2N2905A, H2N2905A					0.44	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$			
M2N2905A, D2N2905A, R2N2905A, H2N2905A					1.75	V dc
Subgroup 2	1019					
Steady-state total dose Irradiation						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO2}			
M2N2905A, D2N2905A R2N2905A, H2N2905A					10	nA dc
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 0.1 mA d	h_{FE1}			
M2N2905A, D2N2905A R2N2905A H2N2905A				75 25 25		
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE2}			
M2N2905A D2N2905A R2N2905A H2N2905A				89 82 43 13	450 450 450 450	
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc	h_{FE3}			
M2N2905A D2N2905A R2N2905A H2N2905A				90 87 57 32		

See footnote at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, pulsed (see 4.5.1)	h_{FE4}			
M2N2905A				84		
D2N2905A				81		
R2N2905A				58		
H2N2905A				48		
Forward current transfer Ratio	3076	V_{CE} = 10 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE5}			
M2N2905A, D2N2905A				50		
R2N2905A				40		
H2N2905A				24		
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B.= 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$			V dc
M2N2905A, D2N2905A					0.44	
R2N2905A					0.44	
H2N2905A					0.50	
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B.= 50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$			V dc
M2N2905A, D2N2905A					1.75	
R2N2905A					2.2	
H2N2905A					2.5	

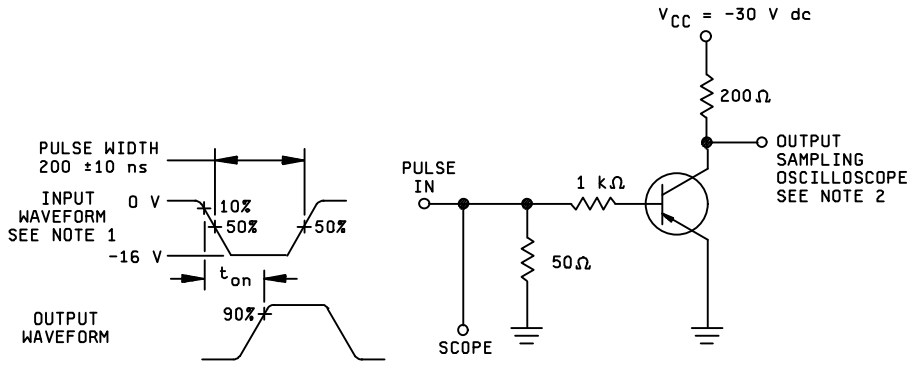
1/ For sampling plan, see MIL-PRF-19500.

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TABLE III. Group E inspection (all quality levels) - For qualification only

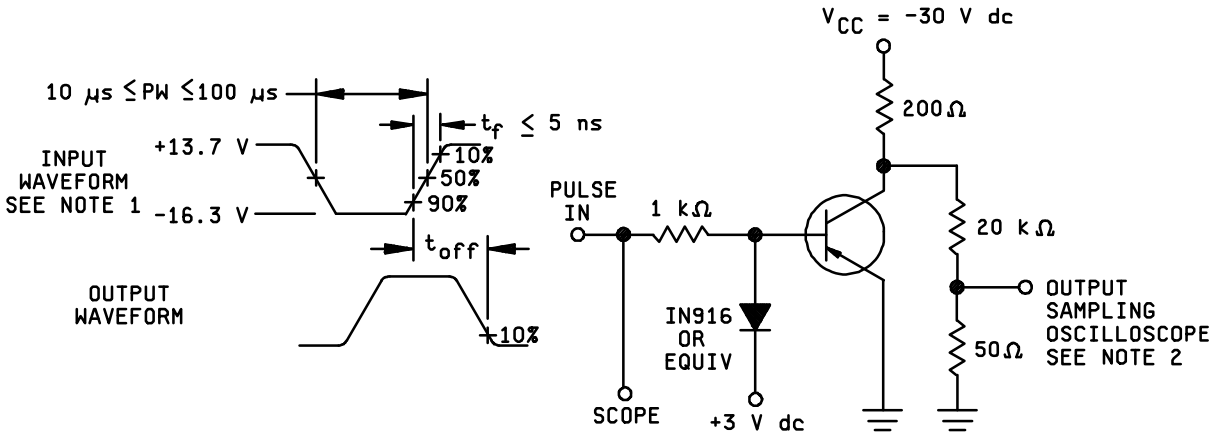
Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See group A, subgroup 2 and 4.5.3 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq +100°C$; forced air cooling allowed on cooling cycle only.	
Electrical measurements		See group A, subgroup 2 and 4.5.3 herein.	
Subgroups 3, 4 and 5			
Not applicable			

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 2. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 3. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation (see 2.2.1).

c. The lead finish as specified (see 3.4.1).

d. Type designation and quality assurance level.

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA – CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 99

Preparing activity:
DLA - CC

(Project 5961-2109)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/290H	2. DOCUMENT DATE 000710

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPES 2N2904, 2N2904A, 2N2904AL, 2N2905, 2N2905A, AND 2N2905AL, JAN, JANTX, JANTXV, JANJ, JANS, JANTXVM, JANTXVD, JANTXVR, JANTXVH, JANSM, JANSD, JANSR, AND JANSH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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